MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Triage for Urinary Tract Infections Treatment in Urology by Nursing/MD/DO/PA/NP Workflow

This workflow is intended for Urology providers seeing established patients with signs or symptoms of uncomplicated Urinary Tract Infections (UTI).

Patient Triage	Antibiotic Flowsheet	Oral Antibiotics	Intravenous Antibiotics

Triage steps for patient calling with signs or symptoms of UTI:

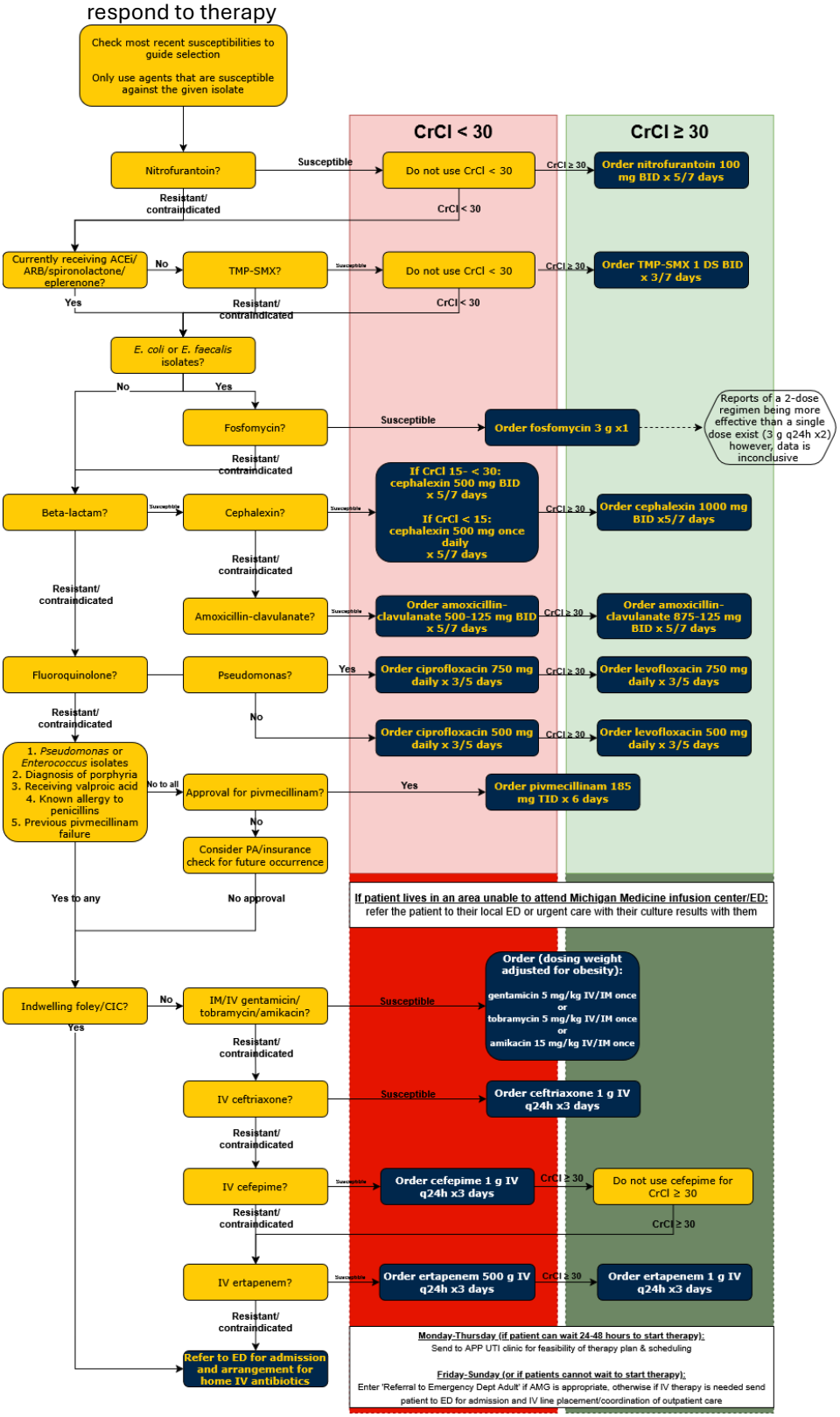
1. If patient is not a current patient in our clinic and has not been seen in 3 years, we cannot provide care.
    a. Direct to local urgent care, their PCP or ER. Can offer NP appt to resume their continued care but will not be in time to treat this episode.
    b. Orders CANNOT be placed for inactive patients.
2. Record symptoms in acute UTI template, order UA with reflex to culture
    a. If patient wants "test of cure" after a recent treatment with no symptoms – this is not appropriate based on our guidelines UNLESS patient undergoing ureteroscopy or stone treatment, penile prosthesis or artificial urinary sphincter in next 30 days or patient is currently pregnant.
    b. If patient wants to discuss UTI prevention or change their current prevention strategy – offer RV either in person or VV with APP as management is not appropriate over phone.
3. Patient with **fever/vomiting/flank pain/inability to take oral liquids/medication or other alarm symptoms (SOB, dizziness/falling…). Empiric abx are not appropriate and will only delay correct care.**
    a. **Send to ED**
4. Patient with **recent urologic surgery within last 30 days** (other than simple endoscopy such as cysto/botox/hydrodistention etc.)
    a. **Contact provider who performed the procedure**
    b. If no available MD/DO in your clinic, contact MD/DO on call.
    c. Likely empiric antibiotics will be ordered once urine sample is submitted.
5. Ensure allergies are correct and up to date
6. Ensure pharmacy is correct and up to date
7. Instructions while waiting for results: **can use dot phrase created below**
    a. *"While awaiting results, you may use the following over-the-counter medications for symptom relief (as long as these are medications you have taken safely in the past) : Phenazopyridine (AZO) 200 mg three times daily after meals for urinary pain, burning, urgency, and frequency (this will turn your urine orange/red). You may also take acetaminophen (Tylenol) or ibuprofen (Motrin/Advil) as directed on the package for pain relief. Drink at least 6-8 glasses of water daily to help flush the urinary tract.*

    The risk of progression to serious complications such as sepsis is less than 1% in otherwise healthy adults with uncomplicated UTIs, and it is safe to wait for culture results to guide appropriate antibiotic therapy.

    Seek immediate medical attention if you develop fever (temperature > 100.4°F/38°C), flank or back pain, nausea, vomiting, shaking chills, worsening symptoms, or inability to tolerate oral fluids."
8. Result: UA negative for nitrites and leukocyte esterase = NEGATIVE for UTI. No treatment and if antibiotic was started they will be instructed to stop.
    a. Culture will not be read, treated or interpreted (even if + since it is perineal contaminant).
    b. Given the negative test patients may want a visit to discuss the symptoms- RV video or in person with APP.
    c. If culture is negative- no treatment same as above.
    d. If culture reveals "GU flora" or contaminants or group B strep or lactobacilli the result is negative (this is your normal biome) same as above.
9. Culture positive with symptoms of UTI (dysuria, urgency, frequency, suprapubic pain, new onset hematuria, autonomic dysreflexia in spinal cord injury) – treated based on flow below and allergies.

TREATMENT OF ACUTE UTI (symptoms limited to the bladder):

- Review past and current urine culture.
- Use short duration option:
    1. For women or men with no retention or catheter use and no recent surgery
- Use long duration option:
    1. For women or men in retention, recent surgery, CIC or foley catheter, or patients who historically are slow to respond to therapy



- **Listed in order of preference for cystitis:**
  Exceptions include recent treatment with antibiotic (within last 2 weeks) and rapid recurrence: choose next option on the list.
  1. Nitrofurantoin (do not use for CrCl < 30)
     - 100 mg BID
     - 5 or 7 days
  2. TMP/SMX (do not use if on ACE/ARB/spironolactone or if CrCl < 30)
     - CrCl ≥ 30: 1 DS tab BID
     - 3 or 7 days
  3. Fosfomycin (*E. coli* or *E. faecalis* ONLY)
     - 3 g once
     - One dose
  4. Cephalexin
     - CrCl ≥ 30: 1000 mg BID; CrCl 15-30: 500 mg BID; CrCl < 15: 500 mg once daily
     - 5 or 7 days
  5. Amoxicillin-clavulanate
     - CrCl ≥ 30: 875-125 mg BID; CrCl < 30 500-125 mg BID
     - 5 or 7 days
  6. Fluoroquinolone
     - If *Pseudomonas*:
       - CrCl ≥ 30: Levofloxacin 750 mg once daily; CrCl < 30: Ciprofloxacin 750 mg once daily
     - Non-Pseudomonas:
       - CrCl ≥ 30: Levofloxacin 500 mg once daily; CrCl < 30: Ciprofloxacin 500 mg once daily
  7. Pivmecillinam (NOT for Pseudomonas or Enterococcus, PCN allergy, on valproic acid, have porphyria, or have prior pivmecillinam failure
     - 185 mg TID
     - 6 days
     - Given likely need for prior authorization, start this process as a tool to use for the *next UTI*, but would proceed to next step to avoid a delay in care.

If no oral option based on allergies or resistance:

1. If patient has allergies (with or without resistance) that preclude oral options to the antibiotics listed above, consider referral to allergy for allergy testing and/or desensitization.
   a. Patients should be informed of referral and that future care will be very challenging and more likely to result in hospital admission unless this is done
   b. On referral, please state – "would like input from Allergy on if allergies to X and Y antibiotic are true allergies, or if they can be used in the future for recurrent UTIs"
2. Can consider gentamicin bladder instillation per protocol 30-60 cc BID x 7 days for patients who already CIC
3. **If no oral options exist: Refer to Urology Rapid UTI clinic for same day appointment to assess if IV therapy in infusion center vs. ED referral are needed we cannot treat a UTI with IV antibiotics via phone**
   a. **If patient lives in an area unable to attend Michigan medicine infusion center:** please refer them to local ER or urgent care and advise them to bring their culture results with them. We cannot provide orders for specific antibiotics for outside facilities; it will be up to the provider they see to make the determination.
   b. If no aminoglycoside resistance, or indwelling foley or CIC: single dose of aminoglycoside
      i. **If Mon-Thurs:** place therapy plan for aminoglycoside (gentamicin, tobramycin, or amikacin) and route to MIST for approval and scheduling. If this can be completed within 48hrs and patient is ok with waiting, proceed with this option. If this cannot be completed in that time, refer to the ED for one time aminoglycoside dose. Place note and a referral (using .UTIEDEXPECTED to communicate the plan with the ED).
         1. APP can order IV abx via "Antibiotics for MDR / Allergy Uncomplicated UTI" therapy plan
            a. Gentamicin 5 mg/kg IV (adjusted for obesity)
            b. Tobramycin 5 mg/kg IV (adjusted for obesity)
            c. Amikacin 15 mg/kg IV (adjusted for obesity)

ii. **If Friday-Sunday:** refer to the ED for one time aminoglycoside dose. Place note and referral (using .UTIEDEXPECTED to communicate the plan with the ED
    1. **Referral to Emergency Dept Adult** (for Michigan Medicine ED only)
    2. .UTIEDEXPECTED - "@NAMELEGAL@ is being referred to the ED to receive a single dose of {aminoglycoside:24265} for uncomplicated UTI. Please see today's note from {ID/Urology:86844} for more detailed information."

c. If resistance to aminoglycosides, if susceptible to ceftriaxone or ertapenem, and eligible for short (3 day) duration:
    i. **If Mon-Thurs:** place therapy plan for 3 days of ceftriaxone or ertapenem and route to MIST for approval and scheduling. This can be done with PIVs daily at MIST and avoid a midline. If this can be completed within 48hrs and patient is ok with waiting, proceed with this option. If this cannot be completed in that time, refer to the ED observation admission for midline placement and home IV antibiotics.
        1. APP can order IV abx via "Antibiotics for MDR / Allergy Uncomplicated UTI" therapy plan
            a. Ceftriaxone
                i. 1 g IV q24h
                ii. 3 days
            b. Cefepime (for CrCl < 30 ONLY)
                i. 1 g IV q24h
                ii. 3 days
            c. Ertapenem
                i. CrCl ≥ 30: 1 g IV q24h; CrCl < 30: 500 mg IV q24h
                ii. 3 days
    ii. **If Friday-Sunday**: refer to the ED for observation admission for midline placement and home IV antibiotics. Place note and referral to communicate the plan with the ED.

d. **If indwelling foley or CIC**, then patient will need midline for ertapenem or cefepime or alternative IV antibiotic for 7 days. Refer to the ED for observation admission for line placement and home IV antibiotic coordination. Place eConsult to ID for review and consideration of future treatment options (including TMP-SMX with lab monitoring, or possible single dose aminoglycoside).

e. **If you are not clear on the best plan, please place E consult to ID**. This is meant to help with recommendations for empiric options for future recurrences. ID will respond within 3 business days, will review cultures, allergies, medications, and will provide guidance for future episodes.

Last revised: 05/2026	**Antimicrobial Stewardship Subcommittee:** 02/2026	**P&T:** N/A

References

	Citation
1	Gupta K, et al. International Clinical Practice Guidelines for the Treatment of Acute Uncomplicated Cystitis and Pyelonephritis in Women: A 2010 Update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. Clin Infect Dis 2011;52:e103-e120.
2	Hooton TM, et al. Diagnosis, Prevention, and Treatment of Catheter- Associated Urinary Tract Infection in Adults: 2009 International Clinical Practice Guidelines from the Infectious Diseases Society of America. Clin Infect Dis 2010;50:625-663.
3	Nelson Z, et al. Guidelines for the Prevention, Diagnosis, and Management of Urinary Tract Infections in Pediatrics and Adults: A Wiki Guidelines Group Consensus Statement. JAMA Network Open. 2024;7;(11):e2444495.
4	Lodise TP, et al. Pivmecillinam for Treatment of Uncomplicated Urinary Tract Infection: New Efficacy Analysis. Clin Infect Dis. 2025 Dec 24;81(5):e285-e293.
5	Ackerman AL, et al. Updates to Recurrent Uncomplicated Urinary Tract Infections in Women: AUA/CUA/SUFU Guideline (2025). J Urol. 2026 Jan;215(1):3-12.

Revision History

Date	Revision

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source